

Angular Learning Roadmap

Phase 1: Foundations (Week 1-2)

Goal: Understand Angular fundamentals, build basic components, and handle forms.

Topics to Cover:

- Angular CLI & Project Setup
- Components: Inputs, Outputs, Events
- Data Binding (1-way, 2-way)
- Built-in Directives (ngIf, ngFor, etc.)
- Pipes (custom and built-in)
- Template-driven forms

Mini Projects:

- Task Tracker App
- Simple Calculator
- Contact Form with Validation

Phase 2: Intermediate Concepts (Week 3-4)

Goal: Learn to structure Angular apps, work with APIs, and understand reactive programming.

Topics to Cover:

- Reactive Forms (FormBuilder, FormGroup)
- Services & Dependency Injection
- Routing & Navigation + Route Guards
- HttpClientModule for REST APIs
- Basic RxJS (Observable, Subject, map, filter, etc.)

Mini Projects:

- Blog App with Routing and API
- Employee Directory using Forms and Search
- Weather App using HTTP and RxJS

Phase 3: Advanced Angular (Week 5-7)

Goal: Master performance, scalability, and reusability.

Topics to Cover:

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- Angular Signals (v16+)
- Change Detection & OnPush Strategy
- Content Projection (ng-content)
- Lazy Loading Modules
- Smart & Dumb Component Architecture
- Advanced RxJS (mergeMap, switchMap, debounceTime)
- Standalone Components (Angular 14+)

Projects:

- Real-Time Search App with Debounce
- Modular E-commerce Product Listing
- Dashboard with Lazy Loaded Routes

Phase 4: Enterprise Skills (Week 8-10)

Goal: Build enterprise-ready applications with testing, SSR, state management, and accessibility.

Topics to Cover:

- State Management (NgRx / SignalStore / Akita)
- Unit Testing (Jasmine, Karma)
- Angular Universal (SSR)
- PWA with Service Workers
- Internationalization (i18n)
- Accessibility (ARIA, keyboard support)
- Modular App Architecture (Core, Shared, Feature modules)
- CI/CD, environment configs

Capstone Projects:

- SaaS-style Dashboard (Admin Panel)
- PWA-based Expense Tracker
- Angular Universal Blog with SSR & Lazy Loading

Optional Enhancements (After Week 10)

- Monorepo Setup using Nx
- Micro Frontends with Module Federation
- Hybrid App with Ionic + Angular

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- Backend Integration (Spring Boot, Node.js)